

Systems of linear equations



1

- a 1
- c 1
- e 0

- b 0
- d Infinite

2

- | | | |
|---|----------------|---------------|
| a | i One solution | ii $(5, -1)$ |
| b | i One solution | ii $(-6, 4)$ |
| c | i One solution | ii $(2, -4)$ |
| d | i One solution | ii $(3, -6)$ |
| e | i One solution | ii $(6, -3)$ |
| f | i One solution | ii $(-4, -2)$ |

3 No because the gradient can be anything.

4

- a To intersect at $(0, 3)$ the line must have a y -intercept of 3, therefore $b = 3$.
- b The value of $m \neq 5$. If $m = 5$ and $b = 3$ then the second line is the same as the first and there is an unlimited number of solutions.

5

- | | |
|-------|------|
| a Yes | b No |
| c No | d No |

Solve systems graphically

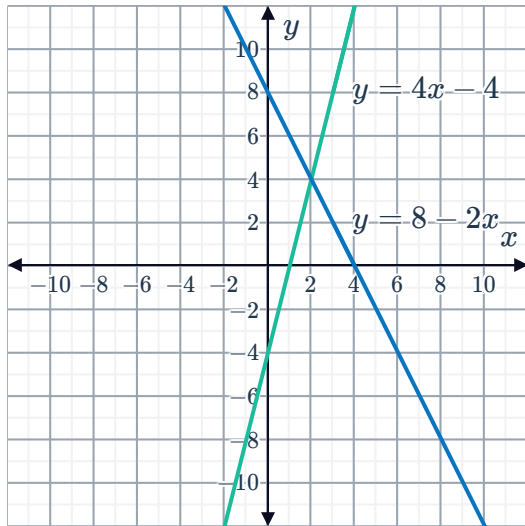
6

- | | | |
|------------------|-------------------|------------------|
| a $x = 6, y = 4$ | b $x = 0, y = -2$ | c $x = 4, y = 0$ |
|------------------|-------------------|------------------|

7

a

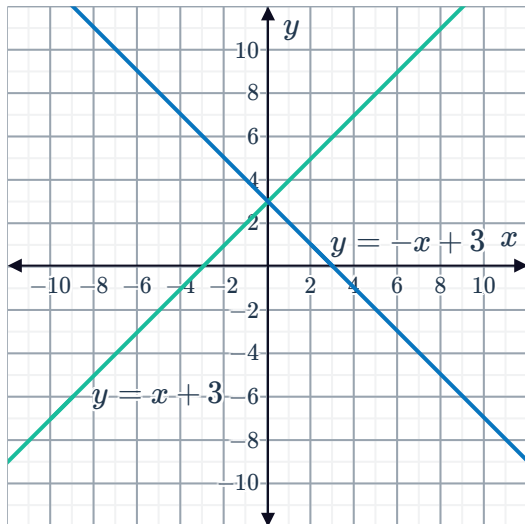
i



ii (2, 4)

c

i

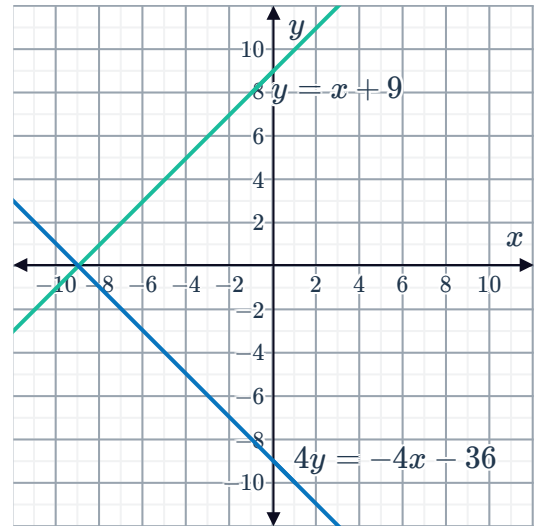


ii (0, 3)



b

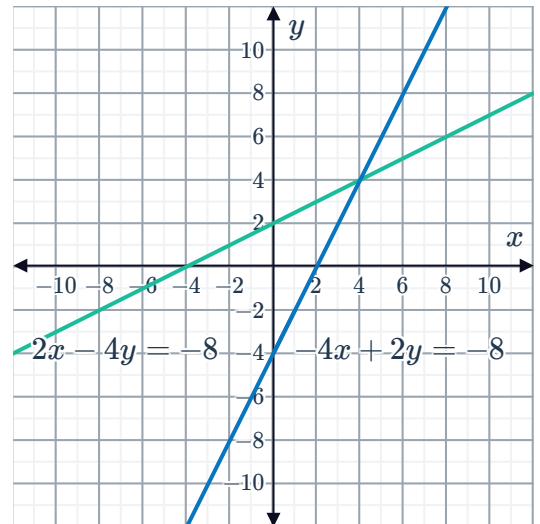
i



ii (-9, 0)

d

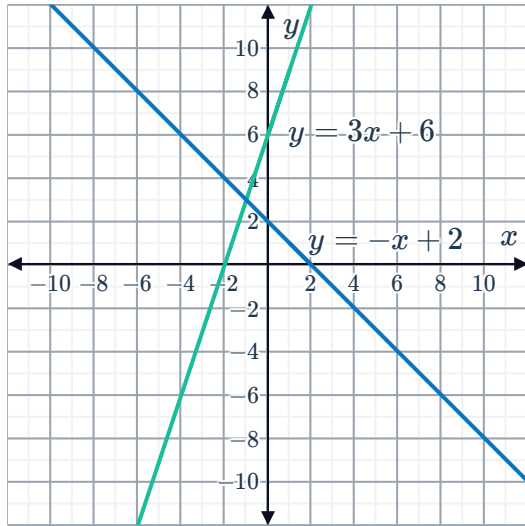
i



ii (4, 4)

e

i

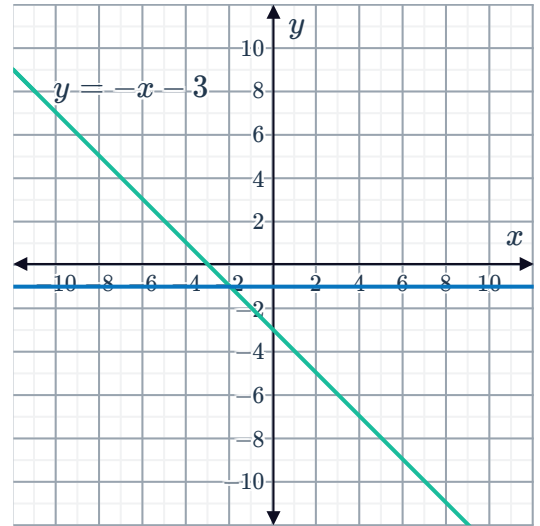


ii $(-1, 3)$



f

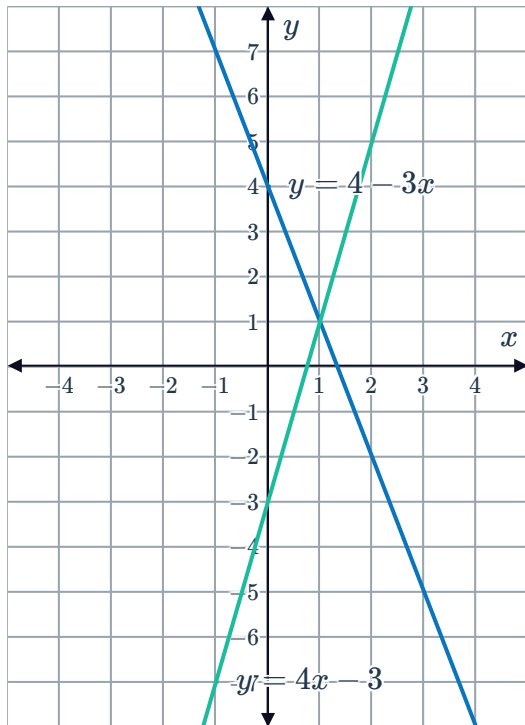
i



ii $(-2, -1)$

g

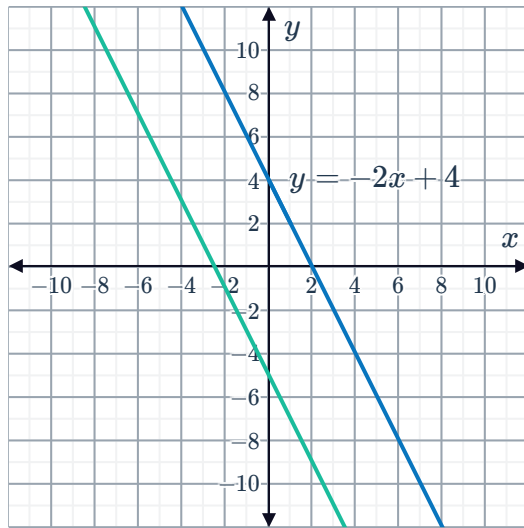
i



ii $(1, 1)$

8

a



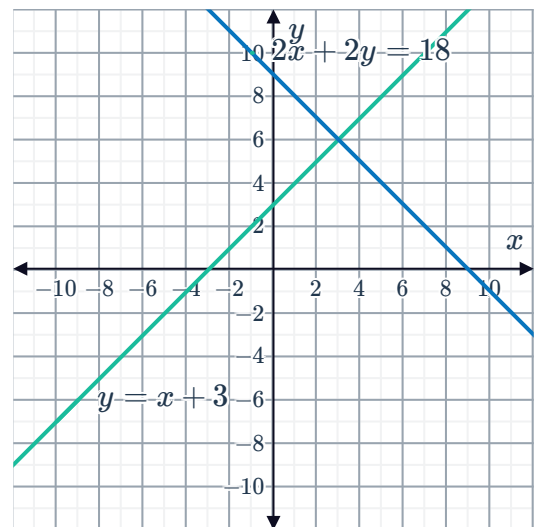
b No, because the lines are parallel.

Applications

9

a $y = x + 3, 2x + 2y = 18$

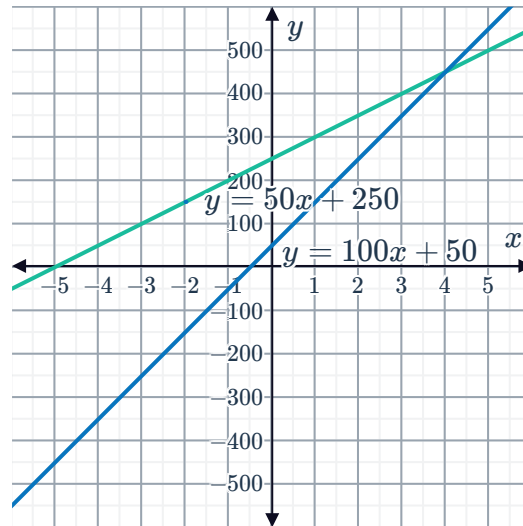
b



c 6 m and 3 m

10

a

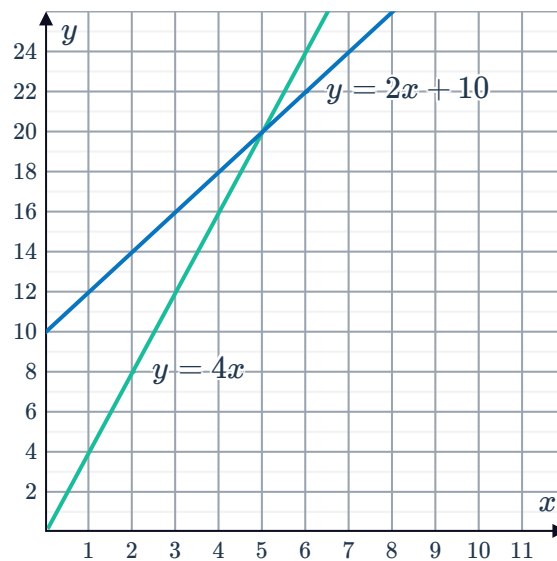


b (4, 450)

c Renting either studio A or B for 4 hours costs the same amount, \$450.

11

a

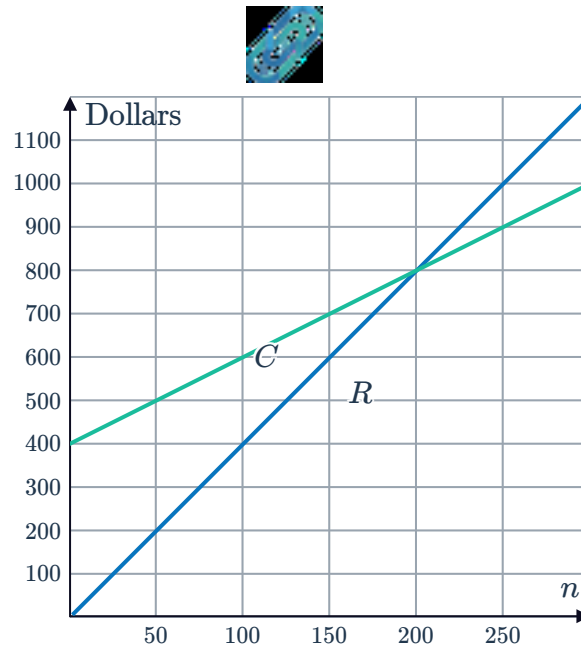


b (5, 20)

c Nonmembers and members pay the same amount per month for taking 5 classes, namely \$20.

12

a



b 200

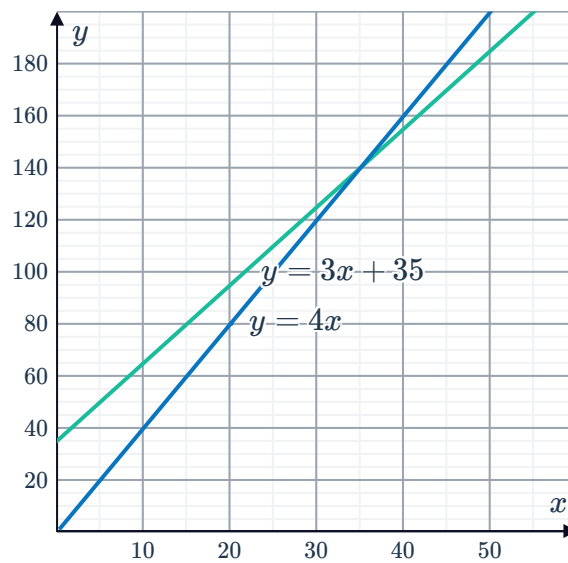
c The amount of revenue generated at the break-even point.

13 (775, 2325)

14

a (35, 140)

b

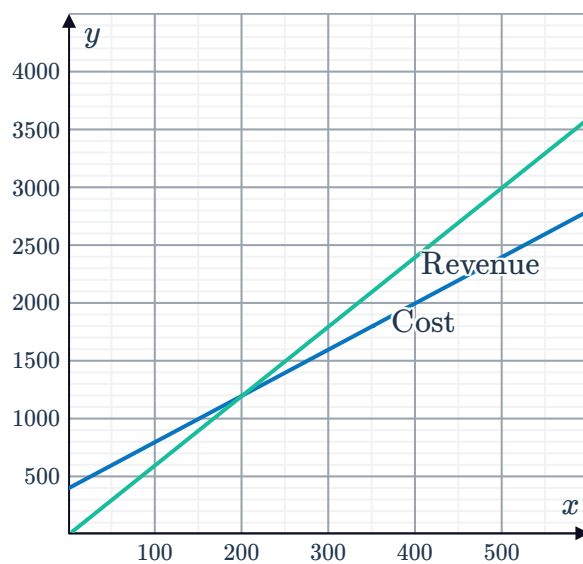


c The point at which Laura is able to just cover her expenses.

15

a (200, 1200)

b



c The company's break even point, at which its revenue is equal to its costs.